

96/100

Problem (1) 14.8

$$a-) J_A = -D_{AB} A_r \frac{dP_A}{dr} = -D_{AB} (2\pi r l) \frac{dP_A}{dr}$$

Assuming steady-state conditions. (30/34)

$$J_A = \text{constant}$$

$$-D_{AB} (2\pi r l) \frac{dP_A}{dr} = C \rightarrow r \frac{dP_A}{dr} = C \quad \checkmark$$

$$\frac{dr}{r} = d(\ln r)$$

$$\cancel{r} \frac{dP_A}{\cancel{r} d(\ln r)} = C \rightarrow \frac{dP_A}{d \ln(r)} = C \quad \checkmark$$

b-) from an excel graph between $\ln(r)$ and wt%.

it's an exponential decay, using EIT, D_{C-Fe}

will be increasing. with the increase of carbon. ✓

(-) wt. %

wt. %	r	$\frac{d \text{ wt. \%}}{dr}$
1.4	4.524	-0.58
1	4.993	-1.08
0.5	5.373	-1.38

$$J_A = -\rho D_{C-Fe} A \frac{d(\text{wt. \%})}{dr}$$

$$\left\{ \begin{array}{l} J_A = \frac{3.6 \times 10^{-3}}{100 \text{ h} (3600 \text{ s/h})} \\ J_A = 1.1 \times 10^{-8} \text{ kg/s} \end{array} \right.$$

$$1.1 \times 10^{-8} = -7730 (D_{C-Fe}) \left(2\pi \left(\frac{4.524}{1000} \right) \left(\frac{100}{1000} \right) (-0.58) \right)$$

$$D_{C-Fe} = 7.85 \times 10^{-10} \text{ m}^2/\text{s} \quad \times \text{ too high}$$

and

$$1.1 \times 10^{-8} = -7730 (D_{C-Fe}) \left(2\pi \left(\frac{4.993}{1000} \right) \left(\frac{100}{1000} \right) (-1.08) \right)$$

$$D_{C-Fe} = 3.82 \times 10^{-10} \text{ m}^2/\text{s} \quad \times \text{ too high}$$

and

$$1.1 \times 10^{-8} = -7730 D_{C-Fe} \left(2\pi \left(\frac{4.524}{1000} \right) \left(\frac{100}{1000} \right) (-1.38) \right)$$

$$D_{C-Fe} = 3.3 \times 10^{-10} \text{ m}^2/\text{s} \quad \times \text{ too high}$$

(2)

Problem (2) 14.10

$$J_H = -C_{DA B} \frac{dX_A}{dx} = -D_{AB} \frac{dC_A}{dx} \quad \text{--- (1)}$$

(33/33)

$$N_A = J_A^* A_c = -D_{AB} A_c \frac{dC_A}{dx}$$

$$N_H \int_0^L \frac{dx}{A_c} = -D_{AB} \int_{x_{A1}}^{x_{A2}} dC_A = D_{AB} (C_{A1} - C_{A2}) = D_{AB} C_{A1}$$

Since $A_c = \pi R^2$

$$R = R_1 + \frac{(R_2 - R_1)x}{L}$$

$$N_H \int_{R_1}^{R_2} \frac{dR}{\pi R^2} \frac{L}{R_2 - R_1} = D_{AB} C_{A1}$$

$$\frac{N_H}{\pi} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \frac{L}{R_2 - R_1} = D_{AB} C_{A1}$$

$$N_A = \frac{\pi D_{AB} C_{A1} R_1 R_2}{L}$$

$$= \frac{\pi (0.15 \times 10^{-9}) (2 \times 10^3) (0.0015) (0.015)}{0.02 \text{ m}}$$

$$N_A = 5.301 \times 10^{-15} \text{ kmol/s.}$$

Problem (3) 149

(33/33)

$$\dot{M}_{A,in} + \dot{M}_{A,g} - \dot{M}_{A,out} = \frac{dM_A}{dt} = M_{A,st}$$

molecular weight.

$$N_{A,r} + N_{A,g} - N_{A,r+dr} = N_{A,st}$$

Since $N_{A,r} = -D_{AB} 4\pi r^2 \frac{\partial C_A}{\partial r}$

$$N_{A,r+dr} = N_{A,r} + \frac{\partial N_{A,r}}{\partial r} dr$$

$$N_{A,g} = N_A (4\pi r^2 dr)$$

$$N_{A,st} = \frac{\partial}{\partial t} [C_A (4\pi r^2 dr)]$$

$$N_A 4\pi r^2 dr + 4\pi \frac{\partial}{\partial r} (D_{AB} r^2 \frac{\partial C_A}{\partial r}) dr = 4\pi r^2 \frac{\partial C_A}{\partial t} dr$$

$$\frac{1}{r^2} \frac{\partial}{\partial t} (D_{AB} r^2 \frac{\partial C_A}{\partial r}) + N_A = \frac{\partial C_A}{\partial t}$$

for spherical
↓

$$\frac{1}{r^2} \frac{\partial}{\partial t} (D_{AB} r^2 \frac{\partial x_A}{\partial r}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial}{\partial \phi} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (D_{AB} \sin \theta \frac{\partial x_A}{\partial \theta}) + N_A = \frac{\partial C_A}{\partial t}$$

(4)